

Exact D.E.s :

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Defn The equation $M(x,y)dx + N(x,y)dy = 0$ is called exact D.E if there exist a function $F(x,y)$ with Continuous partial Derivatives such that $\frac{\partial F}{\partial x} = M(x,y)$ and $\frac{\partial F}{\partial y} = N(x,y)$. Then the general solution is $F(x,y) = C$

Fact: If $M(x,y)$ and $N(x,y)$ have Continuous partial derivatives. Then

$M(x,y)dx + N(x,y)dy = 0$ is exact iff

$$M_y = N_x \quad \text{i.e.} \quad \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

How to find the solution?

$$\textcircled{1} \quad F(x,y) = \int M(x,y)dx + g(y)$$

$$(2) \quad F(x,y) = \int N(x,y) dy + h(x)$$

Ex#1) Solve the IVP :

$$(2xy - 9x^2) + (2y + x^2 + 1) \frac{dy}{dx} = 0, \quad \underline{\text{I.C: } y(0) = 3}$$

Solution) Put the D.E in standard form.
multiply by dx both sides to get:

$$\Rightarrow (2xy - 9x^2)dx + (2y + x^2 + 1)dy = 0$$

$$M(x,y) = 2xy - 9x^2 \quad \text{and} \quad N(x,y) = 2y + x^2 + 1$$

$$M_y = 2x \quad \text{and} \quad N_x = 2x \Rightarrow M_y = N_x. \text{ Hence}$$

exact D.E

To find $F(x,y)$:

$$F(x,y) = \int (2xy - 9x^2)dx = x^2y - 3x^3 + g(y)$$

$$\text{So } \boxed{F(x,y) = x^2y - 3x^3 + g(y)} \quad \text{--- (1)}$$

we need to find $g(y)$.

$$F_y = \cancel{x^2} + g'(y) = N(x,y) = 2y + \cancel{x^2} + 1$$

$$\Rightarrow g'(y) = 2y + 1$$

Integrate both sides

(8)

$$\boxed{g(y) = y^2 + y + k} \quad \text{. substitute in equ (1)}$$

$$F(x, y) = x^2 y - 3x^3 + y^2 + y + k = C_1$$

$$\text{Thus, } y(x^2 + 1) + y^2 - 3x^3 = \boxed{C_1 - k} = C$$

$$\boxed{y(x^2 + 1) + y^2 - 3x^3 = C} \quad \text{--- (2) [general solution]}$$

But $y(0) = -3$. That is, $x=0$, $y=-3$

$$\text{So, } -3(0^2 + 1) + (-3)^2 - 3(0)^3 = C \Rightarrow \boxed{C=6}$$

$$\text{Hence, } y(x^2 + 1) + y^2 - 3x^3 = 6$$

$$\underline{\text{OR}} \quad \boxed{y(x^2 + 1) + y^2 - 3x^3 - 6 = 0} \quad \text{(Actual implicit solution)}$$

Q2) Solve the I.V.P

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$$2xy^2 + 4 = 2(3 - x^2y)y' \quad , \quad \underline{\text{I.C.}}: y(-1) = 8$$

Solution

$$2xy^2 + 4 - 2(3 - x^2y) \frac{dy}{dx} = 0$$

$$(2xy^2 + 4)dx + (2x^2y - 6)dy = 0$$

$$M(x,y) = 2xy^2 + 4 \quad \text{and} \quad N(x,y) = 2x^2y - 6 \Rightarrow$$

$$M_y = 4xy$$

$$N_x = 4xy$$

Hence exact D.E (since $M_y = N_x$)

$$\text{Now, } F(x,y) = \int (2x^2y - 6) dy$$

$$F(x,y) = x^2y^2 - 6y + h(x) \quad \text{--- (1)}$$

we need to find $h(x)$, then.

$$F_x = 2xy^2 + h'(x) = M(x,y) = 2xy^2 + 4$$

$$\Rightarrow h'(x) = 4 \Rightarrow h(x) = 4x, \text{ sub in equ (1):}$$

$$F(x,y) = x^2y^2 - 6y + 4x = C \quad \text{--- (2)}$$

$$\text{But } y(-1) = 8 \Rightarrow x = -1, y = 8$$

$$(-1)^2(64) - 6(8) + 4(-1) = C$$

$$\Rightarrow \boxed{C = 12}$$

Thus, $x^2 y^2 - 6y + 4x = 12$

$$\boxed{x^2 y^2 - 6y + 4x - 12 = 0} \text{ (Actual Implicit Solution)}$$